Changing velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate acceleration and distinguish speeding up, slowing down, and turning.

SUCCESS CRITERIA - I can...

1. Calculate acceleration from velocity change and time.
2. Compare acceleration direction with velocity direction.
3. Explain why turning at steady speed is acceleration.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: motion and direction

(a) During a shuttle run, an athlete runs east at 4 m/s for 6 s, then west at 3 m/s for 4 s. Ignore the brief turn. Find total distance and displacement.
(b) A probe's tracking display reports a direction of 30°. What else must the display specify for that angle to be useful?

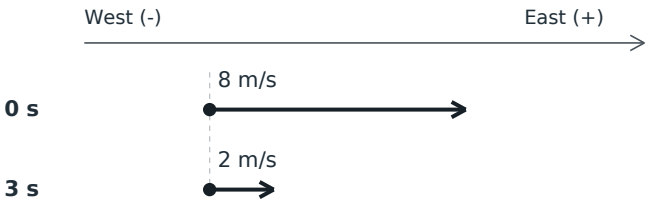
Marzano check - circle: ① ② ③

02 Worked example: braking on a bike

Reference: $a_{avg} = (v_f - v_i)/\Delta t$.
v_i and v_f are initial and final velocity. Δt is elapsed time. With a steady change in velocity, this gives the constant acceleration.

Worked reasoning: A cyclist brakes while riding east. Her speed drops steadily from 8 m/s to 2 m/s in 3 s. Take east as positive.
The velocity change is 2 - 8 = -6 m/s.
Acceleration = -6 ÷ 3 = -2 m/s².
She is still moving east, but her acceleration points west.

Your turn: The model shows the cyclist's velocity before and after braking. Add a separate acceleration arrow below it. Explain why she is slowing down even though both velocity arrows point east.

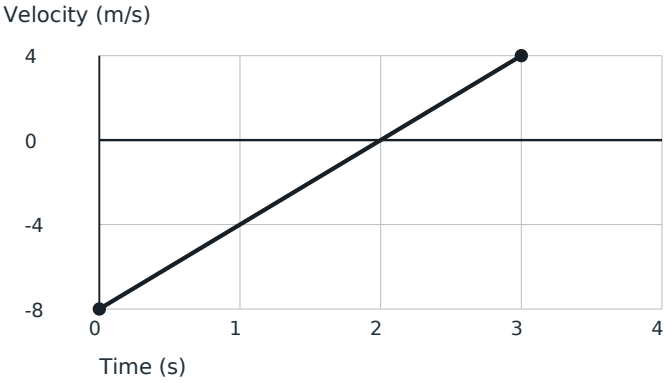


Acceleration: add a separate arrow.

Marzano check - circle: ① ② ③

03 A test car changes direction

The graph shows a remote-control car in a straight-track test. East is positive.
(a) Calculate its acceleration.
(b) When is it momentarily at rest?
(c) Describe its direction of motion and whether it speeds up or slows down before and after that instant.



Marzano check - circle: ① ② ③

04 Braking for stopped traffic

A driver sees stopped traffic ahead. The car slows steadily from 24 m/s to rest in 7.0 s. Start the clock when braking begins, and take forward as positive.
(a) Calculate the acceleration.
(b) How far does the car travel while braking?
(c) Explain why the two answers have different signs.

Marzano check - circle: ① ② ③

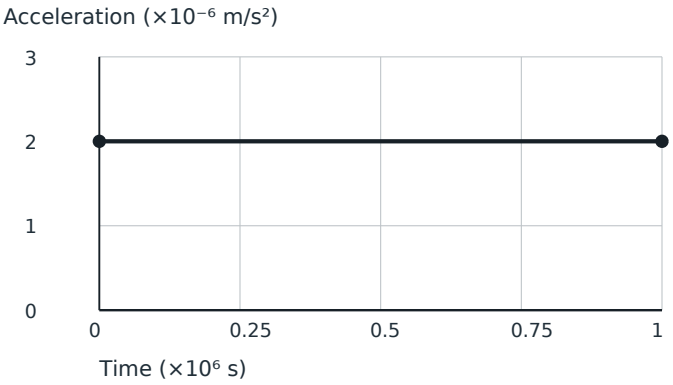
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Continue down each column.

05 Can a tiny thruster matter?

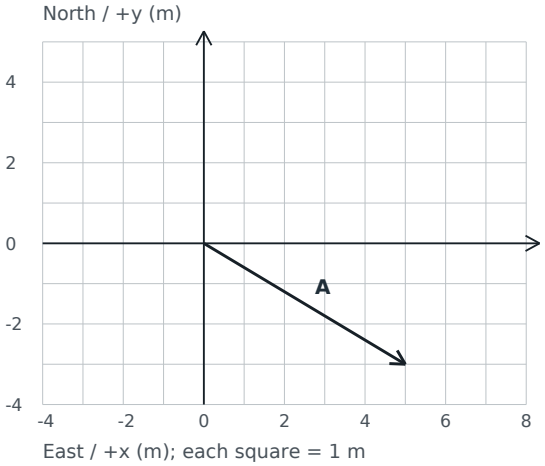
A spacecraft's small thruster produces a constant sideways acceleration. Ignore other sideways forces. The graph uses powers of ten on both axes.
(a) Shade the area under the line and calculate the change in sideways velocity.
(b) Why is the tiny acceleration alone not enough reason to dismiss the thruster?



Marzano check - circle: ① ② ③

06 Review: return the robot to its start

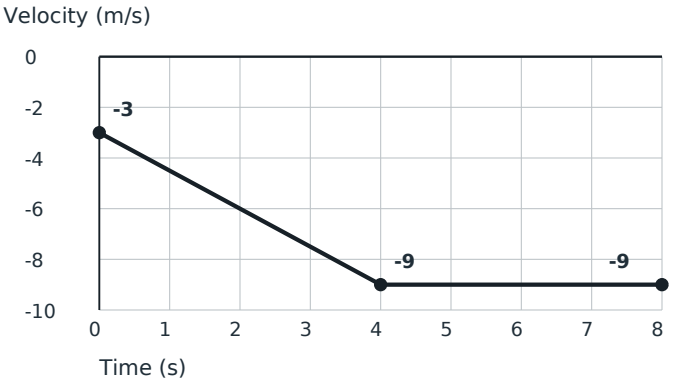
A robot's first move is 5 m east and 3 m south, shown as A. Its second move is 3 m west and 7 m north.
(a) Add the second move head-to-tail on the grid.
(b) Find the resultant displacement: components, magnitude and direction.
(c) Draw the direct return to the start and give its components.



Marzano check - circle: ① ② ③

07 Exit check: thrusters on, then off

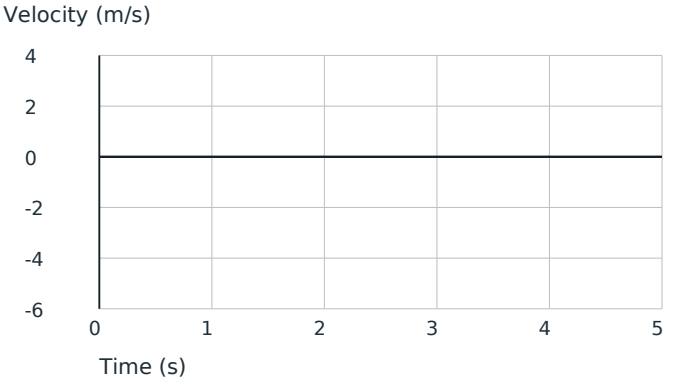
The graph shows a simulated spacecraft with its thrusters on from 0 to 4 s, then off from 4 to 8 s. East is positive. For each interval, calculate acceleration and describe its direction of motion and whether it speeds up, slows down or keeps a steady speed.



Marzano check - circle: ① ② ③

08 STRETCH: past the turning point

A remote-control car initially travels west at 6 m/s. It then accelerates at 2 m/s² east for 5 s, including after it reverses. East is positive.
(a) Complete the graph and mark when it reverses.
(b) Use the signed areas to find displacement and total distance. Explain why they differ.



Marzano check - circle: ① ② ③